

Proposition: Let $T: V \rightarrow W$ be a monomorphism of finite dimensional vector spaces. Then $\dim_F(V) \leq \dim_F(W)$.

Proof: Let $\{v_1, v_2, \dots, v_n\}$ be a basis of V . Then, since $\{v_1, v_2, \dots, v_n\}$ is a linearly independent set, we know that $\{T(v_1), T(v_2), \dots, T(v_n)\}$ is also a linearly independent set. Therefore, $\dim_F(W) \geq n$ (why?)

Proposition Let $T: V \rightarrow W$ be an epimorphism of finite dimensional vector spaces. Then $\dim_F(V) \geq \dim_F(W)$.

Proof Let $\{w_1, w_2, \dots, w_m\}$ be a basis of W . Then since T is an epimorphism, there are elements $v_1, v_2, \dots, v_m \in V$ such that $T(v_1) = w_1, T(v_2) = w_2, \dots, T(v_m) = w_m$. We claim that $\{v_1, v_2, \dots, v_m\}$ is a linearly independent set.

Suppose that $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_m v_m = 0$ for scalars $\alpha_1, \alpha_2, \dots, \alpha_m \in F$. Then

$$\begin{aligned} T(\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_m v_m) &= 0 \\ \implies \alpha_1 w_1 + \alpha_2 w_2 + \dots + \alpha_m w_m &= 0 \end{aligned}$$

$$\Rightarrow \alpha_1 = 0, \alpha_2 = 0, \dots, \alpha_m = 0.$$

Thus $\{v_1, v_2, \dots, v_m\}$ is a linearly independent subset of V .

This implies $\dim_{\mathbb{F}}(V) \geq m$. Thus $\dim_{\mathbb{F}}(V) \geq \dim_{\mathbb{F}}(W)$ $\square$

Let $T: V \rightarrow W$ be a linear transformation of finite dimensional vector spaces.

Recall

① kernel of a linear transformation is a subspace of V .

Therefore, if $v_1, \dots, v_n \in \ker(T)$ then

$$\text{Span}\{v_1, \dots, v_n\} \subset \ker(T).$$

② Image of a linear transformation, that is,
 $\text{im}(T) = \{T(v) : v \in V\}$, is a
 subspace of W .

Some observations.

① If $T(v) = 0$ for some $v \neq 0$ then the line joining v and 0 maps to 0 .

That is, if $L = \{\alpha v : \alpha \in \mathbb{F}\}$ then $T(L) = 0$

② If $T(v) \neq 0$ for some $v \neq 0$ then

$$T(L) = \{ \alpha T(v) : \alpha \in F \}$$

is again a line (line in W that passes through 0 and $T(v)$)

(H.W) ③ If X is a subspace of V then

$$\dim_F(T(X)) \leq \dim_F(X).$$

Proof: Let $\{x_1, x_2, \dots, x_m\}$ be a basis of X . Then

$$\dim_F(X) = m \text{ and } T(X) = \text{Span} \{ T(x_1), T(x_2), \dots, T(x_m) \}.$$

Since every spanning set contains a basis,

$$\dim_F(T(X)) \leq m = \dim_F(X).$$

Linear Transformations & Bases

Theorem Let V and W be finite dimensional F -vector spaces and $\{v_1, \dots, v_n\}$ be a basis of V . Let $w_1, w_2, \dots, w_n$ be n elements of W . Then there is exactly one linear transformation $T: V \longrightarrow W$ such that

$$T(v_i) = w_i \text{ for } i=1, 2, \dots, n.$$

Note: Compare with problem 8, HW-2.

Proof

Define a function $T: V \longrightarrow W$ by

$$T(\alpha_1 v_1 + \dots + \alpha_n v_n) = \alpha_1 w_1 + \dots + \alpha_n w_n.$$

Why this function is a linear transformation?

For any $x, y \in V$, we can write

$$x = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n \text{ and}$$

$$y = \beta_1 v_1 + \beta_2 v_2 + \dots + \beta_n v_n$$

Then

$$\begin{aligned} \textcircled{a} \quad T(x+y) &= T(\alpha_1 v_1 + \dots + \alpha_n v_n + \beta_1 v_1 + \dots + \beta_n v_n) \\ &= T((\alpha_1 + \beta_1) v_1 + (\alpha_2 + \beta_2) v_2 + \dots + (\alpha_n + \beta_n) v_n) \\ &= (\alpha_1 + \beta_1) w_1 + (\alpha_2 + \beta_2) w_2 + \dots + (\alpha_n + \beta_n) w_n \\ &= \alpha_1 w_1 + \dots + \alpha_n w_n + \beta_1 w_1 + \dots + \beta_n w_n \\ &= T(x) + T(y). \end{aligned}$$

$$\begin{aligned} \textcircled{b} \quad T(\alpha x) &= T(\alpha \alpha_1 v_1 + \alpha \alpha_2 v_2 + \dots + \alpha \alpha_n v_n) \\ &= \alpha \alpha_1 w_1 + \dots + \alpha \alpha_n w_n = \alpha (\alpha_1 w_1 + \dots + \alpha_n w_n) \\ &= \alpha T(x). \end{aligned}$$

Hence T is a linear transformation.

Suppose that there is another linear transformation $T_1: V \longrightarrow W$

such that $T_1(v_i) = w_i$ for $i=1, 2, \dots, n$.

$$\begin{aligned}
 \text{Then, } T_1(x) &= T_1(\alpha_1 v_1 + \dots + \alpha_n v_n) \\
 &= \alpha_1 T_1(v_1) + \alpha_2 T_1(v_2) + \dots + \alpha_n T_1(v_n) \\
 &= \alpha_1 w_1 + \alpha_2 w_2 + \dots + \alpha_n w_n \\
 &= T(x), \text{ for any } x \in V.
 \end{aligned}$$

Proposition. Let W_1 and W_2 be subspaces of a vector space V .
Then, $W_1 \cap W_2$ is a subspace of V .

$$\text{Note: } W_1 \cap W_2 = \{w \in V : w \in W_1 \text{ and } w \in W_2\}.$$

Proof

(G1) If $u, v \in W_1 \cap W_2$ then $u, v \in W_1$ and $u, v \in W_2$.

Therefore, $u+v \in W_1$ and $u+v \in W_2$. Thus $u+v \in W_1 \cap W_2$.

(G2) Follows as $W_1 \cap W_2 \subseteq V$.

(G3) $0 \in W_1 \cap W_2$ as $0 \in W_1$ and $0 \in W_2$.

(G4) If $u \in W_1 \cap W_2$ then $-u \in W_1$ and $-u \in W_2$.

Therefore $-u \in W_1 \cap W_2$.

Note that for $\alpha \in F$ and $u \in W_1 \cap W_2$, we have $\alpha u \in W_1$ and $\alpha u \in W_2$. Thus $\alpha u \in W_1 \cap W_2$. Now (V1), (V2), (V3) and (V4) easily holds.